



PRESIDENCY UNIVERSITY

Private University Estd. in Karnataka State by Act No. 41 of 2013
Itgalpura, Rajankunte, Yelahanka, Bengaluru - 560064



**UDYOG SAARTHI APP: PROGRESSIVE
WEB-BASED APPLICATION
A PROJECT REPORT**

Submitted by

ANNIE MARY JOSEPH	20221CCS0040
JOZACH RAJAN SAMUAL	20221CCS0044
KHUSHI YADAV	20221CCS0045

Under the guidance of,

Dr. SRIDEVI S.

BACHELOR OF TECHNOLOGY

IN

**COMPUTER SCIENCE AND ENGINEERING,
(CYBER SECURITY)**

PRESIDENCY UNIVERSITY

BENGALURU

DECEMBER 2025



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PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

BONAFIDE CERTIFICATE

Certified that this report “Udyog Saarthi App: Progressive Web-Based Application” is a bonafide work of “Annie Mary Joseph (20221CCS0040), Jozach Rajan Samual (20221CCS0044), Khushi Yadav (20221CCS0045)”, who have successfully carried out the project work and submitted the report for partial fulfilment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE AND ENGINEERING, INTERNET OF THINGS during 2025-26.


27/11/2025

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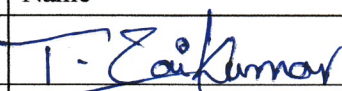
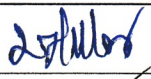
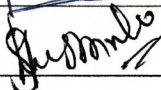
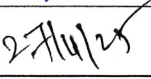


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PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND
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DECLARATION

We the students of final year B.Tech in COMPUTER SCIENCE AND ENGINEERING, INTERNET OF THINGS at Presidency University, Bengaluru, named Annie Mary Joseph, Jozach Rajan Samual, Khushi Yadav, hereby declare that the project work titled “Udyog Saarthi App: Progressive Web-Based Application” has been independently carried out by us and submitted in partial fulfillment for the award of the degree of B.Tech in COMPUTER SCIENCE ENGINEERING, INTERNET OF THINGS during the academic year of 2025-26. Further, the matter embodied in the project has not been submitted previously by anybody for the award of any Degree or Diploma to any other institution.

Annie Mary Joseph USN: 20221CCS0040

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PLACE: BENGALURU

DATE:

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ANNIE MARY JOSEPH
JOZACH RAJAN SAMUAL
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Abstract

Persons with Disabilities (PwDs) in India, especially those with intellectual challenges, continue to face major barriers to employment because information about the 4% government job reservation quota is scattered and often difficult to interpret. Lengthy, jargon-heavy notifications spread across multiple websites leave many positions unfilled and limit workforce participation. This paper introduces Udyog Saarthi, a Progressive Web Application (PWA) developed to address this gap by offering a centralized and accessible source for verified job opportunities.

The paper outlines how the platform's architecture evolved and the practical decisions taken along the way, comparing an early, unstable API-first ingestion method with a more dependable cron-based Python scraping service. A Node.js backend then processes the collected data and uses the Google Gemini API to simplify complex job descriptions into clear, readable text before storing the results in PostgreSQL and serving them through a React interface.

We examine the implementation of several user-focused assistive features, such as a *Guardian Hub for caregiver support*, a *privacy-aware face-recognition login system*, and a *mock-interview module*. The paper also reflects on performance considerations related to advanced accessibility, including our exploration of client-side gesture controls using MediaPipe and the later decision to postpone this feature to maintain responsiveness and stability on low-end devices.

By making practical engineering choices that emphasize reliability and essential functionality, Udyog Saarthi effectively addresses the information gap identified at the outset. The platform reduces cognitive load, supports both users and caregivers, and helps improve the participation of PwDs in the workforce, contributing directly to the goals of the government's reservation mandate.